

Stage: Statistical Characterisation of High-Field Magnet Operations

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- **CNRS** (Centre National de la Recherche Scientifique) [?]
 - Largest public research organisation in France
- **LNCMI** (Laboratoire National des Champs Magnétiques Intenses)
 - National facility for high magnetic field research
 - Based in Grenoble & Toulouse
 - Member of **EMFL** (European Magnetic Field Laboratory)
 - Provides access to high-field facilities

Objectives

- Organise experimental records from LNCMI magnets
- Build a unified database of experiment files
- Compute operational indicators from magnet runs
- Develop tools for data exploration and monitoring

Work Completed Last Week

- Installed and validated MagnetDB analysis tools
- Prepared local copies of experimental records
- Studied the structure of magnet experiment data
- Developed an initial dashboard for experiment exploration

Environment Setup

- Built a Docker-based development environment
- Installed:
 - python_magnetrun
 - python_magnetgeo
 - python_magnetcooling
- Resolved Git submodule issues
- Created isolated Python virtual environment

Dataset Preparation

- Imported experimental Pupitre archives
- Organised local storage of experimental files:
 - `~/LNCMIG-Data/records`
- Verified compatibility with MagnetDB workflows
- Prepared datasets for statistical processing

- For each run, several indicators are computed:

- Total experiment duration:

$$T_{\text{run}} = \sum_i \Delta t_i$$

- Time spent above magnetic field threshold

$$T_{\text{field}} = \sum_{B_i > B_{\text{th}}} \Delta t_i$$

- Electrical energy consumption

$$E_{\text{elec}} = \sum_i P_{\text{tot},i} \Delta t_i$$

- For each run, several indicators are computed:
 - Estimated extracted thermal energy

$$E_{\text{thermal}} = \sum_i (T_{sb,i} - T_{eb,i}) Q_i \rho c_p \Delta t_i$$

where

- T_{eb} : inlet water temperature
- T_{sb} : outlet water temperature
- Q : water flow rate
- ρ : water density
- c_p : specific heat capacity

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127.0.0.1

MagnetDB Dashboard

Experiments: 755
Processed: 434

id	name	site_name	energy_J	heat_extracted_J	duration_s	duration_field_on_s	status
311	2025.07.19 - 17:56:45	M10_A250429_00	0	3981340646.11	2760	42	STATS DONE
312	2025.07.19 - 18:43:02	M10_A250429_00	97087.17	121048877555.67	16300	14833	STATS DONE
313	2025.07.20 - 11:55:07	M10_A250429_00	114717.52	144173349218.89	19448	15494	STATS DONE
314	2025.07.20 - 19:49:47	M10_A250429_00	16290.43	20259663029.67	2582	2261	STATS DONE
315	2025.07.20 - 22:12:55	M10_A250429_00	21027.48	26436389398.61	3597	3127	STATS DONE
316	2025.07.21 - 12:15:37	M10_A250429_00	141723.77	181928585641.83	27717	19516	STATS DONE
317	2025.07.22 - 15:30:17	M10_A250429_00	0	347600207.5	181	54	STATS DONE
318	2025.07.22 - 20:36:54	M10_A250429_00	86989.81	109606347370.72	15633	13249	STATS DONE
319	2025.07.23 - 13:45:56	M10_A250429_00	14559.91	20402163423.06	4378	3820	STATS DONE
320	2025.07.23 - 14:59:14	M10_A250429_00	242652.19	297097220511	36552	33296	STATS DONE
321	2025.07.24 - 13:56:14	M10_A250429_00	190522.39	239711713134.83	34950	28034	STATS DONE
322	2025.07.25 - 19:26:27	M10_A250429_00	18809.96	26748133609.17	5449	3018	STATS DONE
323	2025.07.26 - 18:26:46	M10_A250429_00	88052.19	115395839902.78	20087	15355	STATS DONE
324	2025.07.27 - 20:29:26	M10_A250429_00					pending
325	2025.07.28 - 12:39:31	M10_A250429_00					pending
326	2025.08.29 - 15:36:07	M10_A250429_00					pending
327	2025.09.10 - 11:50:41	M10_A250429_00					pending
328	2025.09.10 - 17:14:45	M10_A250429_00					pending
329	2025.09.11 - 11:55:36	M10_A250429_00					pending
330	2025.09.11 - 19:15:32	M10_A250429_00					pending

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- Experimental datasets available locally
- MagnetDB Python tools operational
- Statistics computation pipeline validated
- Dashboard connected to experiment records
- Ready for large-scale statistical processing

Possible Future Work

- Complete integration of experimental statistics into MagnetDB workflows
- Automatic computation of statistics for newly imported experiment files
- Secondary database to store processing results without modifying the original database